

Data Backup and Recovery Standard Operating Procedure

This SOP sets the step-by-step standard for backup scheduling, validation, restore testing, and controlled production recovery.

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1.0 Purpose

This SOP ensures that backups are not only created, but also verified, retained, and recoverable within the business expectations for critical systems. In a production-level RAG dataset, the procedure must answer practical recovery questions such as what is backed up, how restore confidence is established, and what steps must be followed before service is reopened.

The procedure supports both resilience and governance. It defines a repeatable operating model so teams can prove backups exist, detect failures early, and execute recovery with enough discipline to protect data integrity during stressful incidents.

2.0 Scope

This SOP applies to production databases, document stores, file repositories containing regulated or business-critical information, and configuration assets required to restore service. Teams must maintain a current inventory of systems covered by the backup program.

The procedure covers backup schedule design, monitoring, validation, restore testing, production recovery, and recovery record retention. Service-specific runbooks may add product detail, but they should not weaken the controls defined here.

- Applies to critical data stores, file systems, and infrastructure configuration assets.
- Requires encryption, retention rules, and restore testing for systems in scope.
- Uses backup logs, monitoring, and recovery records as the auditable source of truth.

3.0 Responsibilities

The responsibilities below define who must act, approve, review, and maintain records so the process can be executed consistently across teams and audit periods.

3.1 Infrastructure Operations

- Maintain schedules, retention settings, monitoring, and validation routines for systems in scope.
- Investigate failed backups, lead restore drills, and coordinate controlled recoveries during incidents.
- Document recovery timelines, restore points, and exception decisions.

3.2 System Owners

- Confirm data classification, recovery objectives, and business-critical dependencies for their systems.
- Participate in restore testing and validate that recovered data is operationally usable.
- Review backup coverage when architecture, schemas, or storage patterns change.

3.3 Incident or Change Leads

- Approve production recovery actions during incidents or recovery tests.
- Coordinate communication, access control, and reopening of systems after validation.
- Ensure follow-up actions are assigned when backup or recovery gaps are discovered.

4.0 Backup Management Procedure

Backup activity should follow an explicit schedule matched to recovery point objectives and data criticality. A backup that technically exists but does not meet the business recovery target or is not being monitored effectively should be treated as a control gap rather than a success.

4.1 Step 4.1 Define and Run Backup Schedule

- Set daily incremental and weekly full backups for critical systems unless a stricter requirement applies.
- Use encryption, approved storage targets, and retention settings aligned to policy.
- Verify that schedule ownership and monitoring are assigned for each covered system.

4.2 Step 4.2 Monitor Backup Completion

Backup jobs should be observed through alerts and periodic review rather than assumed successful. Repeated failures, long runtimes, or missing off-site copies indicate a resilience issue that must be addressed before a real recovery event exposes the weakness.

- Route backup failures and missed jobs to the operations queue.
- Investigate and resolve repeated failures within the same business day.
- Confirm off-site and retention controls remain active after configuration changes.

5.0 Validation and Restore Testing Procedure

Restore confidence comes from evidence, not assumptions. Regular restore drills should be scheduled so teams can confirm that backups are complete, usable, and consistent with expected recovery times. Validation should include both technical integrity and application-level usefulness where possible.

5.1 Step 5.1 Run Restore Drill

- Select the backup set and target restore point for the scheduled validation.
- Restore into a controlled environment when time and risk profile allow.
- Capture timing, errors, and manual workarounds required to complete the restore.

5.2 Step 5.2 Validate Recovered Data

System owners should confirm that recovered data is complete enough for business use and that the application or reporting layer behaves as expected. Technical success without operational usability should still be treated as a failed or incomplete validation.

- Check schema integrity, record counts, key workflows, and essential reference data.
- Document discrepancies between expected and recovered state.
- Open follow-up actions for missing automation, unclear runbooks, or slow recovery steps.

6.0 Production Recovery Procedure

When a real recovery is required, the team should confirm the business objective, chosen restore point, and acceptable data-loss window before taking action. Recovery work must be controlled because restoring the wrong point or reopening a system too early can create a second incident on top of the original problem.

6.1 Step 6.1 Approve Restore Point and Start Recovery

- Confirm the selected restore point with the incident lead and system owner.
- Record the expected data-loss window and rationale for the chosen point.
- Begin restoration using the approved runbook and note each major action.

6.2 Step 6.2 Validate Before Reopening

- Check data completeness, application behavior, and security controls before enabling user access.
- Confirm dependent systems or downstream jobs will not reintroduce corruption or inconsistency.
- Reopen service only after the designated approver confirms validation is sufficient.

7.0 Exceptions and Escalation

Any system in scope that cannot meet the backup or restore control sequence requires a documented exception, owner, risk statement, and remediation plan. Teams should not normalize weak recovery posture by treating it as a temporary inconvenience without time-bound follow-up.

Escalate immediately if backup failures persist, if restore drills reveal unusable data, or if recovery timing materially exceeds the agreed objective. Backup confidence erodes quickly when gaps are known but not acted on.

8.0 Records and Review

Backup logs, restore drill outcomes, exception records, and real recovery timelines must be stored in approved operational systems. These records support audit, resilience review, and faster recovery the next time an event occurs.

Infrastructure Operations reviews backup coverage and recovery performance quarterly and after any major recovery event. Repeated manual recovery steps should trigger automation or runbook improvements rather than repeated acceptance of the same risk.